Google Cloud AI Builder Cup · Disaster management

ResilienceOps AI

One command view for a district disaster: AI agents that forecast, prepare, dispatch and track relief — and a person who approves every step.

Built for DDMA Wayanad, Kerala · demo event WYN-2026-001, landslide & flash flood (simulated)

The problem

In the first hours of a landslide, the district decides with scattered, late, unverified information

Forecasts don't become plans

An IMD red alert reaches the control room, but turning it into "who goes where by when" is manual, and the window before peak rain is a few hours.

Reports arrive in pieces

Malayalam voice notes, ward-officer calls, police and gauges arrive separately. Confirming one incident takes calls and minutes.

Quiet places get missed

A ward whose phone tower is down sends few reports, so it looks fine. Relief follows the loudest signal, not the greatest need.

Benefit to the community: faster, verified decisions before and during a disaster, relief that reaches the people who can't call, and a record the district learns from.

Prepare, Respond and Recover on one map, with the next decision always in view

Prepare

Rank places by risk, draft readiness tasks with owners and deadlines, close capacity gaps with reviewed requests, then hand over to Respond.

Respond

Corroborate reports into one incident, recommend who to send and by which route, track every team, and hand over to Recover.

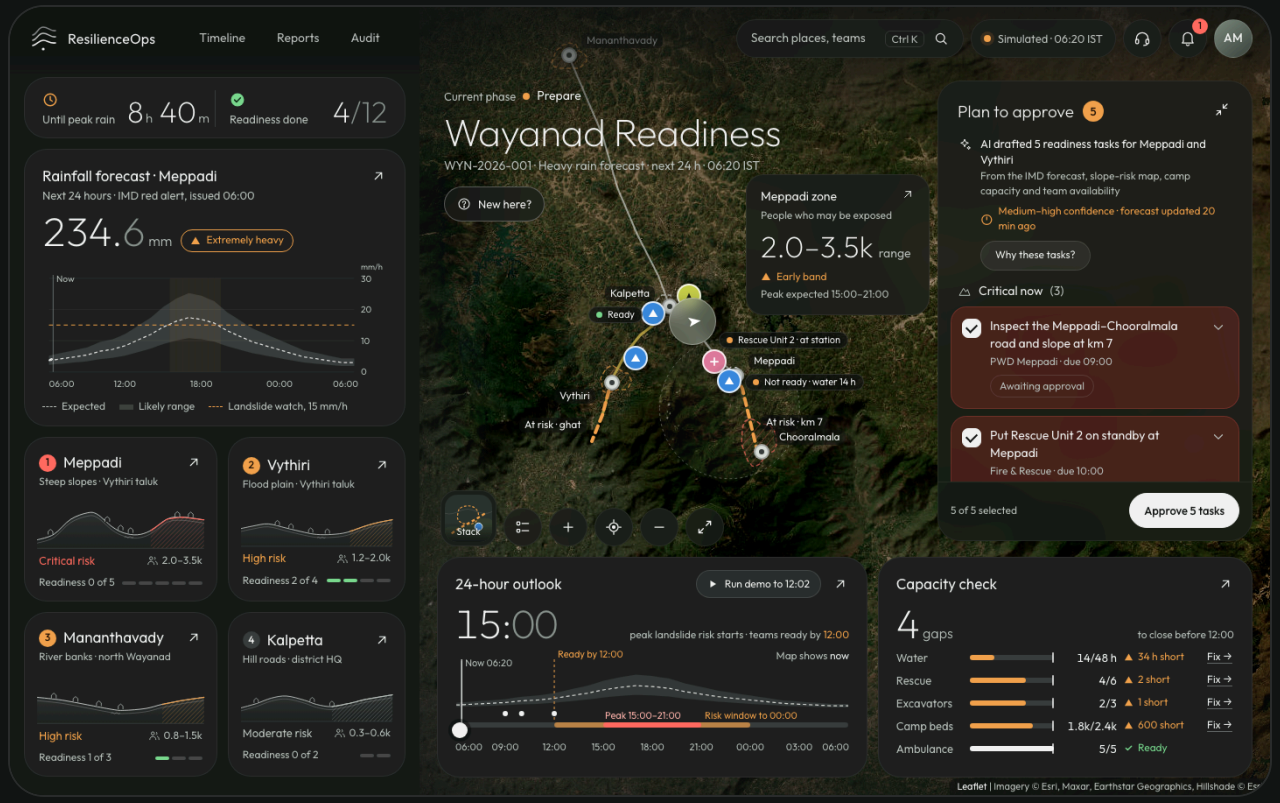
Recover

Count only verified relief, flag communities that may be missed, track restoration and turn what went wrong into next year's plan.

Plus an event timeline, situation reports for KSDMA, an audit trail of every AI recommendation, and a phone app for field teams (English / മലയാളം).

A red alert becomes an approved plan in minutes

- Forecast & Impact Agent ranks Meppadi critical and says why
- Preparedness Agent drafts 5 tasks with owner, deadline and proof
- Capacity gaps close through reviewed requests (KSDMA, PWD, partners)
- Activating Response hands everything over



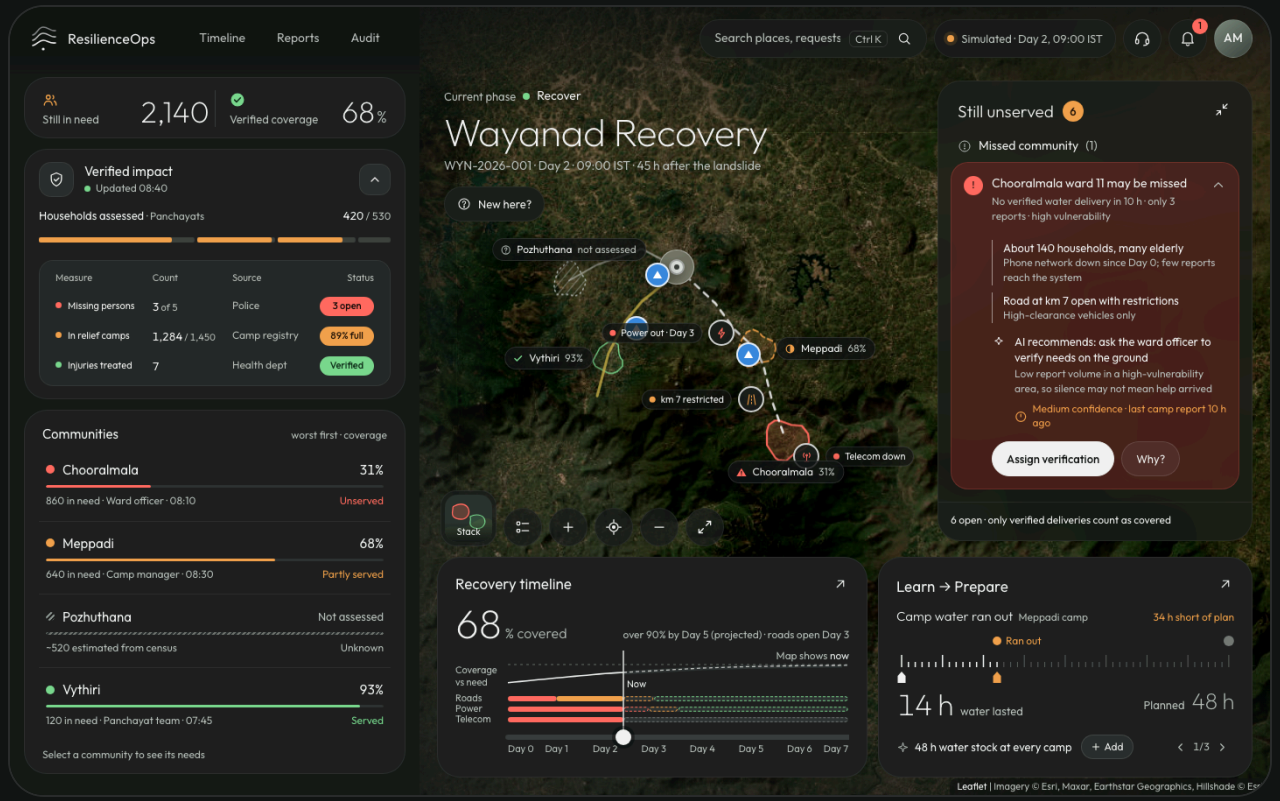
One verified incident, one recommendation, one approval

- Field Intelligence Agent merges a Malayalam voice note, a call and the police into one critical incident
- Dispatch Agent recommends Rescue Unit 2 by Route C, 11 min away because Prepare staged it
- Nothing moves until the Incident Commander approves



Silence is not safety

- Relief Coverage Agent flags Chooralmala: 40% of need, 5% of reports, tower down
- Verify on the ground, approve tankers, count it only when delivery is verified
- Reconciliation Agent turns "water ran out at 14 h" into a Prepare task



11 agents on Google ADK + Gemini, deployed on Cloud Run

Scenario data
Forecast, places, teams, camps, reports (one JSON source)

Tools
Deterministic Python: risk scores, team matching, coverage gaps

Gemini via ADK
The agent calls its tools, then writes the why, the drafts and the confidence

FastAPI on Cloud Run
One service: the API and the app; results cached, Regenerate on demand

A person decides
Approvals go to the decision log and the timeline

Prepare	Forecast & Impact · Risk Memory · Preparedness	Respond	Field Intelligence · Dispatch & Delegation · Forecast vs Ground Reality
Recover	Relief Coverage · Damage Assessment · Recovery & Reconciliation	All	Communications (English + Malayalam) · Command Orchestrator

Every AI output is a recommendation with its evidence

Recommend, never act

Dispatch, requests, advisories, reports and phase changes all need an approval. Analysts and coordinators recommend to the Incident Commander.

Show the evidence

Each recommendation carries its sources, verified / estimated / reported tags, confidence and limitation, and lands in the Audit trail.

No invented numbers

Figures come from tools, not the model. No casualty figures, ever. Ranges stay ranges.

Works when things fail

If Gemini or the network is down, the app keeps its built-in text and an offline map. WCAG 2.2 AA throughout.

Built on what districts already have; grows by data, not code

Feasible now

Public inputs (IMD, CWC, KSDMA layers), the DEOC's own registers and phones. Runs in a browser; field teams use a phone page in their language. One Cloud Run service.

Scales across districts

A new district is a new scenario file: places, teams, camps, routes. Agents, rules and UI stay the same. Cloud Run scales to zero between events and out during one; Firestore keeps the record.

Next

Live IMD/CWC feeds, the fleet register, GIS road layers, WhatsApp intake for voice notes, KSDMA reporting format.

Honest scope

The demo event is simulated. Place names are real; events and figures are illustrative.

Thank you

Forecast → prepare → respond → recover → learn

Live app: [Cloud Run URL] · Code: [GitHub URL] · Demo: [video URL]